## Wall time per step: Splat simulation, 1M (964,763 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

| | bixverse mtx | bixverse h5ad | bixverse GPU* | bixverse MLX* | Scanpy mtx | Scanpy h5ad | Scanpy dask | scrapper mtx | scrapper h5ad | Seurat std mtx | Seurat std h5ad | Seurat BPCells |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| load | 97 | 72 | 57 | 44 | OOM | OOM | 39 | OOM | OOM | OOM | OOM | 133 |
| qc | 0.5 | 1.2 | 0.4 | 0.4 | – | – | 7.2 | – | – | – | – | 9.8 |
| normalise | n/a | n/a | n/a | n/a | – | – | 0.0 | – | – | – | – | 4.3 |
| hvg | 1.4 | 1.6 | 1.2 | 1.1 | – | – | 8.2 | – | – | – | – | 25 |
| pca | 6.6 | 6.8 | 2.3 | 3.1 | – | – | 18 | – | – | – | – | 468 |
| harmony | 3.1 | 3.0 | 1.9 | 1.2 | – | – | 4.2 | – | – | – | – | 179 |
| knn_snn | 29 | 29 | 15 | 13 | – | – | 40 | – | – | – | – | 100 |
| cluster | 8.0 | 8.4 | 6.5 | 6.2 | – | – | 47 | – | – | – | – | 367 |
| cluster_fast | 45 | 46 | 8.0 | 6.2 | n/a | n/a | n/a | – | – | n/a | n/a | n/a |
| umap | 79 | 78 | 10 | 9.8 | – | – | 555 | – | – | – | – | 555 |
| dge | 3.2 | 3.3 | 5.4 | 2.9 | – | – | – | – | – | – | – | 822 |
| tsne_default | 641 | 643 | 525 | 526 | – | – | TIMEOUT | n/a | n/a | – | – | TIMEOUT |
| tsne_fft | 195 | 193 | 29 | 45 | – | – | – | n/a | n/a | – | – | – |
| tsne_quick | 136 | 134 | 72 | 73 | n/a | n/a | n/a | – | – | n/a | n/a | n/a |
| total (no tSNE) | 227 | 204 | 100 | 81 | failed | failed | 719 | failed | failed | failed | failed | 2,662 |

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.